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B.Tech. VII - Sem. (Main/Back) Examination, January - 2023
Computer Science And Engineering
7CS4-01 Internet of Things

Time : 3 Hours

Maximum Marks : 120
Min. Passing Marks : 42

Instructions to Candidates:

*Attempt **All Ten** questions from **Part A**, Five questions out of **Seven** from **Part B** and **Four** questions out of **Five** from **Part C**.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

PART - A

(Answer should be given upto 25 words only)

ALL questions are compulsory. **(10×2=20)**

1. Define the Internet of Things.
2. What is the use of communication APIs in IoT?
3. What is the difference between Wireless Sensor Network and IoT?
4. Compare the boards Arduino Uno and Raspberry Pi with respect to its support to speed, memory and storage.
5. How soil moisture sensor works? Give two applications of soil moisture sensor.
6. What is Uniform Resource Identifier?
7. List any four security challenges in IoT.
8. What are the differences between IoT and M2M communication?
9. What is Software Defined Network?
10. How IoT is useful in the smart cities?

PART - B

(Analytical/Problem solving questions)

Attempt any FIVE questions. **(5×8=40)**

1. Describe the types of various IoT enabled technologies.
2. Explain the two application layer protocols that are used in IoT networks.

3. Explain with a diagram the interfacing pins for input and output on Raspberry Pi board.
4. Design an IoT application using humidity sensor.
5. Describe what the API is. To better understand how Rest and Restful APIs function, please give an example that is both appropriate and comprehensible.
6. What are the features of Routing Protocol for Low Power and Lossy Networks (RPL)?
7. Design a smart fridge using IoT technologies.

PART - C

(Descriptive/Analytical/Problem Solving/Design questions)

Attempt any FOUR questions. **(4×15=60)**

1. What distinguishes the Physical design of the Internet of Things from its Logical counterpart? Explain.
2. Describe the basic components of IoT-A reference model.
3. What is semantic interoperability? Why do we need it?
4. Describe network functions virtualization.
5. Explain the various areas in the agriculture domain that IoT is making an impact.

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B.Tech. VII Sem. (Main) Examination December- 2023
Computer Sc. and Engg.
7CS4-01 Internet of Things

Time : 3 Hours **Maximum Marks : 70**

Instructions to Candidates:

*Attempt **all ten** questions from **Part A**, **five** questions out of **Seven** questions from **Part B** and **Three** questions out of **Five** questions from **Part C**.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205).

PART - A

(Answer should be given up to **25** words only)

All questions are **compulsory.** **(10×2=20)**

1. Define the challenges of IoT. (2)
2. Write the difference between IoT and M2M communication. (2)
3. What is a smart home? (2)
4. What is software defined network? (2)
5. Explain uniform resource identifiers (URIS). (2)
6. List any four design and development challenges in IoT. (2)
7. Explain IoT applications in Environment. (2)
8. Differentiate between Arduino and Raspberry Pi? (2)
9. List layers of IoT protocol. (2)
10. What is the connection between IoT and sensors in the commercial enterprises? (2)

PART – B

[5×4=20]

(Analytical/Problem solving questions)

Attempt any five questions

Q.1 Explain the major services used in cloud computing technology.

Q.2 Describe how wireless sensor networks become one of the enabling technologies of IoT?

Q.3 Explain the differences between Machines in M2M and Things in IoT?

Q.4 Describe the structure of Network Function Virtualization for IoT.

Q.5 What impacts will the Internet of Things (IOT) have on transportation sector?

Q.6 Write down the differences between Rest API and Web Socket API.

Q.7 Describe the steps for designing IoT system with neat diagram.

PART – C

[3×10=30]

(Descriptive/Analytical/Problem Solving/Design Questions)

Attempt any three questions

Q.1 Explain the role of big data analysis in IoT and define its characteristics.

Q.2 Explain how IoT technology used to enable the agricultural industry to increase operational efficiency, lower costs, reduce waste and improve the quality of their field?

Q.3 Describe the implementation of IoT technology into distributed energy system to optimize the efficiency of energy infrastructure and reduce wastage in the renewable energy systems.

Q.4 Explain functional view, information view, deployment and operational view and other relevant architectural views of IoT reference architecture.

Q.5 Write short note on -

(i) Lite OS (ii) Contiki OS (iii) Ultra Sonic Sensor

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B. Tech. VII - Sem. (Main / Back) Exam., - 2024
Computer Science & Engineering
7CS4-01 Internet of Things

Time: 3 Hours **Maximum Marks: 70**

Instructions to Candidates:

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used /calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART – A [10×2=20]

(Answer should be given up to 25 words only)

All questions are compulsory

Q.1 Explain the role of things in IoT.

Q.2 What is the use of GPIO Pins in an IoT device?

Q.3 Explain the flavors of Linux OS supported by Raspberry Pi device.

Q.4 What is the use of SPI interface on Raspberry Pi?

Q.5 What are 4 aspects in one's business to Master IoT?

Q.6 What role does the network play in the Internet of everything?

Q.7 Live few examples of impact of Internet of Things (IoT) on our lives.

Q.8 What are interrupts in Arduino?

Q.9 What Python libraries used in Raspberry Pi to control GPIO Pins?

Q.10 What is application of IoT in Healthcare?

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PART-B

Analytical/Problem Solving questions.

Attempt any Five questions. **(5×8=40)**

1. Why security required in IOT? Explain detail various security in internet of things. **(3+5)**
2. Explain layered IOT architecture using a figure. **(8)**
3. Explain the application of IOT in home automation systems. **(8)**
4. How Big Data can be used in IOT? **(8)**
5. Explain the concepts and terminology of RFID Application. **(8)**
6. What are basic components of sensor network? **(8)**
7. What is threat analysis in internet of things ? Explain in detail threat analysis. **(3+8)**

PART - C

(Descriptive/Analytical/Problem Solving/ Design questions) **(4×15=60)**

Attempt Any Four Questions.

1. What is Role of the IOT for increased Autonomy and Agility in collaborative production Environments ? Write in detail business model scenario for IOT? **(7+8)**
2. Explain in detail clustering principal in IOT? Also write what is software defined Network in detail? **(8+7)**
3. Write in detail business model Scenario for IOT? What are the components and communication media required for making smart building. **(6+9)**
4. Discuss area of development and standardization in IOT. Explain in detail HLSA IOT frame work. **(8+7)**
5. Identify the key characteristics of M2M data. Also explain the data generation, data acquisition, data validation steps, data storage, data processing and data Analysis steps in M2M data management . **(5+10)**

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B.Tech. VII-Sem. (Back) Examination, December - 2023
Computer Sc. & Engg.
7CS4-01 Internet of Things

Time : 3 Hours **Maximum Marks : 120**
Min. Passing Marks : 42

Instructions to Candidates:

*Attempt all **Ten** questions from Part A, Five question out of seven from Part B and Fo9ur questions out of Five from Part C,*

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Use of following supporting material is permitted during examination. (Mentioned in form No. 205).

PART - A

Answer should be given up to 25 words only.

All questions are compulsory. **(10×2=20)**

1. Write the difference between sensors and actuators.
2. What is MQTT?
3. Define smart agriculture.
4. What is IOT connectivity?
5. Explain different characteristics of IOT?
6. What is requirement of IOT protocol standardization?
7. Why do IOT system have to be self adapting and self configuring?
8. Which of the communication protocols are used by IOT?
9. Differentiate between Raspberry pi and desktop computer?
10. Define RFID.

PART - B

(Analytical/Problem solving questions)

Attempt any Five questions. **(5×4=20)**

1. Explain the various areas in the health and lifestyles domain that IoT is making an impact. **(4)**

2. Describe network function virtualization. **(4)**

3. What is cloud computing? Is cloud computing used in Internet of things? Explain the advantages of cloud computing. **(4)**

4. Explain four stages of IoT architecture. **(4)**

5. Write the differences between URI and URL. **(4)**

6. Explain about IoT communication APIS in detail. **(4)**

7. With the help of proper diagram explain the logical design of IoT. **(4)**

PART - C

(Descriptive/Analytical/Problem Solving/Design questions)

Attempt any Three questions. **(3×10=30)**

1. what is wireless sensor network? Explain the IoT levels in details. **(10)**

2. What is sensor? How sensors are different from actuator? Explain humidity sensor, ultrasonic sensor and temperature sensor in detail. **(10)**

3. What is REST model? Why it is important? Explain the REST methods? **(10)**

4. How software defined network different from traditional network? Explain the SDN architecture? What are the four key characteristics of an SDN architecture? **(10)**

5. What do you understand by home automation? What are the basic requirement for home automation? Explain the smart cities build with the help of IoT in detail. **(10)**